Bt401 Final Addition (2022)

Final term Current Paper warrior group of Zoologist

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1. **Botanical garden**

It is a garden dedicated to the collection, cultivation and display of a wide range of plants labeled with their botanical names. It may contain specialist plant collections such as tropical plants, or other species of plants.

2. **forest genetic resources,**

Forest genetic resources are essential for forest-depending communities who rely for a substantial part of their livelihoods on timber and non-timber forest products (forexample fruits, gums and resins) for food security, domestic use and income generation.

3. **outbreeding and inbreeding**

Inbreeding:

"Inbreeding, the mating of individuals or organisms that are closely related through common ancestry."

Outbreeding:

"The intentional breeding of distantly related or unrelated individuals for the purpose of producing offspring of superior quality.

4. **cryopreservation ..**

Cryopreservation (Latin Kuos means frost) means storage of materials at very low temperature. Plant cells and tissue cultures are brought to zero state of metabolism by subjecting them to ultra-low temperature i.e. -I96°C. It is done by using liquid nitrogen which provides approximately -496°C. Cryoprotectants (e.g. glycerol, proline, mannitol,dimethylsulfoxide, sorbitol) are also used to protect the viable cells from the damage during freezing and thawing (to become unfrozen or warm).

5. **Who is responsible for botanical garden**

Botanical gardens are often run by universities or other scientific research organizations, andoften have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science

6. **Germplasm**

Germplasm are living genetic resources such as seeds or tissues that are maintained for the purpose of animal and plant breeding, preservation, and other research uses. ... Germplasm collections can range from collections of wild species to elite, domesticated breeding lines that have undergone extensive human selection.

7. Cold storage 5

Cold storage - it involves storage in low and non freezing temperature.

8. Slow Freezing- Slow thawing

With this method organs are labeled into vials after equilibration with a cryoprotectant solution and then cooled at rate of 0.5-2 oC per minute down to -1 oC. Seeding is then induced and a holding period of 5 to 15 minutes allows equilibration of the temperature. Thereafter embryos are cooled to -60 oC or lower at a rate of 0.3 to 0.5 oC per minute before beingtransfer to liquid nitrogen. Frozen embryos must be slowly thawed at a rate of less then oC per minute to prevent osmotic shock

9. Rapid cooling and rapid thawing

In this technique , however cooling is terminated at -30 to -40 oC and embryos are then plunged into liquid nitrogen for rapid cooling to -196 oC. Thawing is therefore performed
rapidly (200 to 500 oC per minute) to prevent recrystallization

10. Vetrification

Vetrification is the process of cooling where the water in the tissue becomes glass rather than crystals. Glass is a liquid that is too cold (too viscous) to flow. In other words verification is solidification due to increased viscosity rather than crystallization.

11. Ultra rapid Freezing

In this technique serial equilibration of embryos in high concentration of DMSO (3-5 M) supplemented with sucrose (0.3 to 0.5 M). The embryos are then plunged into liquid Nitrogen. Thawing is then done with warm water bath (approximately 500 oC perminute)

12. Conservation role of botanical gardens 5

A botanical garden or botanic garden is a garden dedicated to the collection, cultivation and display of a wide range of plants labelled with their botanical names.
Types of plants in botanical gardens
1.cacti and succulent plants.
2. herb gardens.
3. greenhouses, shade houses.
4. tropical plants.
5. Medicinal Plants.
6. aromatic or textile plants
7. other exotic plants. science.

Importance of Botanical garden

1- Enjoyment 2- Economic 3- Scientific research

Botanic gardens contain collections of plants for education, scientific purposes and display;they can be: taxonomically-based - collections of a particular family, genus or group ofcultivars; or collections of native plants; or useful species such as medicinal, aromatic ortextile plants.

4- Conservation

Conservation of rare and threatened plants. The conservation of plant diversity is critical for sustainable development and botanic gardens are playing a key role as centers of conservation action. Botanical gardens can promote diversity. Because they include many species of plant.

5- Climate Change

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole.

*transpiration in plants can increase water vapour in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. *Contribute to soil fertility and prevent soil erosion

13. Application for permit to import plant genetic resource 5???

(a) the kind and quantity of plant or plant material;

(b) the country and locality of origin;

(c) destination;

(d) the name and address of the consignor and the consignee;

(e) means of transport; (f) the prescribed port or point of entry;

(g) the prescribed for which the plant or plant material is proposed to be imported e.g., consumption propagation or processing.

14. Advantgaes and disadvantages of outbreeding 5

Advantages of Out breeding: • Out breeding often produces offspring of superior quality because it increases

homozygosity (the occurrence of two alleles for the same trait at corresponding positions on homologous

chromosomes) • Sharply reduce the risk of deleterious recessive genes being expressed • One of the benefits

of out breeding is less chance of genetic abnormalities • The ability to make a breed stronger

Disadvantages of Out breeding: outbreeding species. 3 Can destroy well-adapted genotypes because the

offspring (genotypes) from sexual reproduction are not guaranteed to be viable, as is the case with selfing.

15. Protected areas

Definition: Protected areas set aside to protect a specific natural monument. They are generally quite small protected areas and often have high visitor value.

Primary objective To protect specific outstanding natural features. Their associated biodiversity and habitats

16. Domestic animals

Domesticated animals are animals that have been selectively bred and genetically adapted over generations to live alongside humans. They are genetically distinct from their wild ancestors or cousins.

Forexample: dog, cat, goad, cow

17. AnGR threats....?

Despite the importance of animal genetic resources and their diversity, their diversity has been continually decreasing over time.

• One of the greatest threats to livestock diversity is pressure from large-scale commercial production systems
to maintain only high-output breeds.

• Changes in climate will have an impact on livestock and food production in many ways.

• Some major disease threats that livestock currently face include, rinderpest, foot and mouth disease, and Peste des petits ruminants (PPR), also known as sheep and goat plague

18. Applications of germplasm conservation techniques

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed too readily as a basic tool in maintaining genetic diversity.

Conventional seed storage offers several advantages:

1. Seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation 2. Good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes; 3. Seed storage is a relatively cheap method for conserving a broad range of germplasm 4. Large land areas are not tied up in conservation 5. International exchange of genetic material is facilitated by seed storage.

How to conserve plant biodiversity

conservation of plant biodiversity is an important issue concerning the human population worldwide. Conservation of plant biodiversity can be performed in situ and ex situ. These two methods are complementary and are not exclusive. They offer different alternatives for conservation, but selection of the appropriate strategy should

be based on a number of criteria, including the biological nature of the species and the feasibility of applying the chosen methods 22 . At present biotechnological methods have been used to conserve endangered, rare crop ornamental, medicinal and forest species for short- ,medium-, and long- term

19. Biodegradation

Biodegradation is **the degradation of the materials into environmentally acceptable products such as water, carbon dioxide, and biomass by the action of naturally available microorganisms under normal environmental conditions**.

20. Phenotypic plasticity and its role in environmental change

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions." A degree of phenotypicplasticity is found in most trees, but varies substantially amongst and within species.Even in species with very little genetic diversity, such as PinuspineaL, strong phenotypic plasticity is expressed for growth related traits, which may have helped the species colonise new environments

phenotypes under different environmental conditions. The concept is often edžtended to populations and species. with plastic trees those with flexible morphology and physiology that grow at least reasonably well under a range of different environmental stresses without genetic change. A degree of phenotypic plasticity is found in most trees, but varies substantially amongst and within species. Even in species with very little genetic diversity, such as PinuspineaL, strong phenotypic plasticity is expressed for growth-related traits,which may have helped the species colonise new environments. At least in the short term, high plasticity is likely to favour tree survival under changing environmental conditions,although trade-offs between traits can be expected. Since phenotypic plasticity has a heritable basis and may be selected for under changing environments, complex interactions between traits are possible, depending on the magnitude and structure of change.

21. Role of earthworms

earthworms help to maintain soil structure and the availability of water throughout the soilprofile. Studies have shown that the presence of these animals can help to alleviate the effectsof drought on crop Production. Studies have also revealed the remarkable ability of diverse soilinvertebrate communities to restore the structure of degraded soil. He potential for managingsoil invertebrates to enhance their beneficial roles has been little explored

22. Mission of Ramsar convention

The conservation and wise use of all wetlands through local and national actions and international cooperation, as a contribution towards achieving sustainable development

throughout the world.

23. How plant cope effect climate change

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow.

transpiration in plants can increase water vapor in the atmosphere, causing more precipitation and cloud cover in an area.

The additional cloud cover often reinforces the cooling by blocking sunlight.

*Contribute to soil fertility and prevent soil erosion

24. Social culture value of national monument

Category III areas are likely to hold socio-cultural values as they may have features such as sacred groves, springs, waterfalls, mountains, sea coves etc. of importance to one or more faith groups. These areas are often of significant tourism value and can be managed with the objective of promoting sustainable tourism

25. Anthropogenic features

26. Name labe of hosts plant genetic resources

Plant Genetic Resources Institute hosts the sole National Genebank of Pakistan for conservation of plant genetic resources and six labs including

1. germplasm exploration lab
2. seed preservation lab
3. in vitro conservation lab
4. germplasm evaluation lab
5. plant introduction and seed health lab
6. Data management lab

27. Write three pillars of Ramsar conventions

The "three pillars" of the convention

The Contracting Parties (160) commit to:

• Work towards the wise use of all their wetlands.

• Designate suitable wetlands for the list of Wetlands of International Importance.

• Cooperate internationally on trans boundary wetlands, shared wetland systems and shared species.

28. What are the steps of Allotropic speciation?

The biological equivalent is "allopatric speciation," an evolutionary process in which one species divides into two because the original homogenous population has become separated and both groups diverge from each other.

Often this type of speciation occurs in three steps.

- First, the populations become physically separated, often by a long, slow geological process like an uplift of land, the movement of a glacier, or formation of a body of water.
- Next, the separated populations diverge, through changes in mating tactics or use of their habitat.
- Third, they become reproductively separated such that they cannot interbreed and exchange genes.

29. What is the historical background of Ramsar conversation

History: Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was
adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December
1975

30. What is PkU

Phenylketonuria (PKU)
It is a human genetic condition caused by mutations to a gene coding for a particular liver
enzyme. In the absence of this enzyme, an amino acid known as phenylalanine does not get converted into the next amino acid in a biochemical pathway, and therefore too much
phenylalanine passes into the blood and other tissues. Change in environment (lowering Phenylalanine consumption) can affect the phenotype of a particular trait, demonstrating a gene-environment interaction

31. Effect of gene frequecy???

Genetic adaptations that make a population more suited for survival are achieved through gene frequency changes across generation. Many tree species have high genetic variability in adaptive traits and can therefore grow under a wide range of conditions.

32. Name of step of plan genetic conservative

The Global Strategy for Plant Conservation
The Global Strategy for Plant Conservation (GSPC) is a program of the UN's Convention onBiological Diversity founded in 1999. It is a Plan to Save the World's Plant Species - grew out of the Convention on Biological Diversity and is being fed into government policy around the world.
Vision of GSPC: -
Without plants, there is no life. The functioning of the planet, and our survival,

depends on plants. The Strategy seeks to halt the continuing loss of plant diversity"

Objectives:

The GSPC highlights the importance of plants and the ecosystem services they provide for all life on earth, and aims to ensure their conservation. The GSPC has 5 mainobjectives:

1. Plant diversity is well understood, documented and recognized
2. Plant diversity is urgently and effectively conserved
3. Plant diversity is used in a sustainable and equitable manner
4. Plant diversity is used in a sustainable and equitable manner
5. The capacities and public engagement necessary to implement the strategy have been
developed.

33. Zoo garden def

zoo, also called zoological garden or zoological park, **place where wild animals** and, in some instances, domesticated animals are exhibited in captivity. In such an establishment, animals can generally be given more intensive care than is possible in nature reserves or sanctuaries.

34. objectives of Cartagena protocol 5

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force onSeptember 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009

Objectives of Cartagena Protocol on Biosafety

• Contribute to ensuring an adequate level of protection in the field of the safe transfer, handling and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity,

• It takes into account risks to human health, and specifically focusing on transboundary movements.

• It seeks to protect biodiversity from the potential risks of living modified organisms (LMOs) resulting from modern biotechnology

35. Historical back ground of CBD 3

CBD is about the conservation and wise use of different biological resources (plants and animals).

• It was adopted in 1992 at Rio De Janeiro, Brazil and entered force on January,

• 1993, which was 90 days after the 30th ratification.

• Pakistan signed it in June 1992 at United Nations Conference on Environment and Development held at Rio De Janeiro, Brazil

- Pakistan ratified it on 26th July 1994 **Category vi law rules..???**

36. Effect of extreme weather on invertebrate genetic resources

Because of the many ecosystem services that they provide, invertebrates have a key role to play in adapting agriculture to the effects of climate change. The extent to which the individual services provided by invertebrates will be enhanced or impeded by climate change is difficult to predict. However, if invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish.

Healthy soils – and healthy, diverse soil invertebrate communities – will be vital to climate change adaptation. For example, earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop Production. Studies have also revealed the remarkable.

ability of diverse soil invertebrate communities to restore the structure of degraded soil. He potential for managing soil invertebrates to enhance their beneficial roles has been little explored. Few if any deliberate attempts have been made to introduce soil invertebrates into new countries or ecosystems. Given the potential for such species to become invasive, it is inadvisable to attempt any such introductions until soil ecology is much better understood than it is today. However, every effort should be made to avoid agricultural practices that disrupt resident soil invertebrate communities and the services they provide.

It is likely that some pests, as they move into new areas in response to climate change, will at least temporarilyescapefrom their natural enemies. This is likely to increase demand for classical biological control agents, especially in places where the newly established pest population is separated from its original home by a physical barrier such as the sea or a mountain range. For this reason, access to new classical biological control agents are likely to be particularly important for island countries

37. Threatened species of Rannkutch 3

The site supports many locally and globally threatened species, including the great Indian bustard, houbara bustard, sarus crane, and striped hyena and supports more than 1% of the biogeographical population of flamingos

38. What areas are covered by Cartagena protocol?

The Bio safety (Protocol to CBD) deals with;

Safe handling,

Storage

Trans-boundary movement of the Genetically Modified Organisms (GMO).Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009

Objectives of Cartagena Protocol on Biosafety

The Protocol states that it aims to;

• Contribute to ensuring an adequate level of protection in the field of the safe transfer, handling and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity,

• It takes into account risks to human health, and specifically focusing on transboundary movements.

• It seeks to protect biodiversity from the potential risks of living modified organisms (LMOs) resulting from modern biotechnology.

• Cartagena Protocol areas

• The Protocol covers:

• Transboundary movement, transit, handling and use of all living modified organisms thatmay have adverse effects on the conservation and sustainable use of biological diversity,taking into account risks to human health. It does not cover:

• •Products derived from LMOs (e.g. paper from GM trees)

•LMOs, which are pharmaceuticals for humans that are addressed by other relevant international agreements or organizations.

39. . Differentiate between genetic drift and genetic flow. 5

Gene Flow VS Genetic Drift

Comparison Chart in an Image Format

Characteristics	Gene Flow	Genetic Drift
Definition	Process of alleles going from one population to another	Alteration of allele frequency in a gene pool
Cause of Gene Alteration	Migration or geographical isolation	Random sampling with two mechanisms (bottle effect or founder effect)
Cause of Movement	Founder Effect: breaking away of a few members and forming their own group	Transferring of organisms to an already existing population
Other Names or Terms	Gene Migration and Allele Flow	Allelic Drift and Sewall Wright Effect
Chances of Genetic Variation	Most likely ensures genetic variation	Less likely ensures genetic variation as it is mainly due to luck

DB Difference Between.net

40. Define assisted migration .2

Assisted migration is the act of moving plants or animals to a different habitat. The destination habitat may or may not have once previously held the species; the only requirement is the destination habitat must provide the bioclimatic requirements to support the species.

41. 7. Role of pollinator 3

Birds, bats, bees, butterflies, beetles, and other small mammals that pollinate plants are responsible for bringing us one out of every three bites of food. They also sustain our ecosystems and produce our natural resources by helping plants reproduce.

42. Biodiversity of habitat species management 5

Category IV areas are important for their role in 'plugging the gaps' in conservation strategies by protecting key species or habitats in ecosystems. It provides a management

approach for areas that have already undergone substantial modification, necessitating protection of remaining fragments for identified target species with or without intervention

43. Define food security. 2

"Food security [is] a situation that exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life".

44. Habitat species that affects climate change 3

Global warming resulting from human emissions of greenhouse gases. The consequencesinclude habitat loss; shifts in climatic conditions and in habitats that surpass emigrationalcapabilities; altered competitive relationships.

45. The wild bird and animal protection act

An Act to make better provision for the protection and preservationof certain wild birds and animals.

(1) This Act may be called the Wild Birds and Animals Protection Act, 1912; and (2) It extends to the whole of Pakistan.

46. Hingol national park declared reserved in ....

Hingol National Park spread over an area of about 1,650 square km along the Makran Coast,

Balochistan

- It is the largest of National Parks of Pakistan
- The area was for the first time declared reserved in 1988.
- Hingol is known to support threatened invertebrates in addition to a variety of birdspecies
- The park is an excellent habitat to wild animals including over 3000 ibexes, and 1500 Urialsand more than 1200 Chinkara
- A number of resident and migratory birds are supported by this park.

47. In how many ways bacteria can transfer their DNA Horizontally.

There are three mechanisms of horizontal gene transfer in bacteria:

transformation,

Transformation is the genetic alteration of bacteria by direct uptake and incorporation of exogenous DNA from their surroundings.

transduction,

transduction is the process by which a virus transfers genetic material from one bacterium to another.

conjugation.

DNA is transferred between bacteria through a tube between cells. Transposable elements are chunks of DNA that "jump" from one place to another.

48. Silent features of west Pak Fisheries Ordinance 1961.

It is an ordinance to amend and consolidate the law relating to fisheries in the West Pakistan.

• It says that no person will be allowed to use dynamite, pesticides or other explosives for catching the fish.

• Government can declare any water area as sanctuary.

• Then there will be no permit for fishing except some license.

• An Inspector of Fisheries may without a warrant arrest any person committing

• in his view any offence under section 6, 7, 8, 9 or 11

49. Ramsar sites of Punjab.

Uchhali Complex

• Taunsa Barrage

• Chashma Barrage

50. Who established Bahawalpur zoo(2)

Nawab of former state Bahawalpur, Sir Sadiq Muhammad Khan Abbasi, established Bahawalpur Zoo in 1942 . It covers an area of 25 acres.

51. Bottleneck effect (2)

Allele frequencies in a group may be very different from those of the population prior to the event,. Even some alleles may be missing entirely.

The smaller population will also be more susceptible to the effects of genetic drift for generations (until its numbers return to normal).

52. Conservation of FGR.(3)

Forest genetic resources (FGR) are the heritable materials maintained within and among treeand other woody plant species that are of actual or potential economic,environmental, scientificor societal value.

53. Pakistan terrestrial water Maritime zoo act 1976(5)

Pakistan Terrestrial Water & Maritime Zones Act,1976;

• It includes provisions for preservation, development and protection of marine environment

• This law controls marine pollution and exploration, development, conservationand management of living resources in Pakistan’s Exclusive Economic Zone (EEZ)

• This law means that a ship carrying nuclear and hazardous substances willhave to inform the Government of Pakistan.

54. Link between climate change and botanical gardens?(5)

Ans: plants can alter the temperature of the Earth’s atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. *transpiration

in plants can increase water vapor in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight.

*Contribute to soil fertility and prevent soil erosion

55. Law and legislation

Laws The systeŵ of rules which a particular country or community recognizes as regulating the actions of its members and which it may enforce by the impossible penalties.

legislation

The process of making or enacting laws.

56. What is 2nd national park of Pakistan

Kirthar National Park is the second largest national park of Pakistan spread over an area of 3000 square kilometers.

57. Run kutch wild life sanctuary

It spread over 566,375 ha is part of the great Thar desert and comprises. Rann of Kutch across the frontier with India, which includes permanent saline marshes, coastal brackish lagoons, tidal mudflats, and estuarine habitats. The site supports many locally and globally threatened species, Threatened species include the Great Indian bustard, Houbara bustard, Sarus crane this area used to have the only population of the Indian Wild Ass or Onager in Pakistan

58. Tissue culture conservation

Conservation in tissue culture in in vitro genebanks is often combined with cryopreservation. Cultures in the active genebank are maintained by successive subculturing allowing culture renewal and distribution. For medium term storage, sub-culture intervals are extended, reducing processing costs by arresting growth using cold treatments, adapted light conditions, culture medium modifications (osmotic active compounds, growth retardants). This increases efficient use of resources and staff time and offsets selection risks and contamination

59. Note on Indus dolphin

The Indus River dolphin is one of the world's rarest mammals

• It is second most endangered freshwater river dolphin

• Approximately 1,100 specimens of this species exist today in a small fraction of their former range

• Population of this species has gradually declined due to various factors e.g. water pollution, poaching, fragmentation of habitat due to barrages and dolphin stranding in the irrigation canals

60. Wild plant resources

Some of the common wild plant genetic resources are as follow;

- Prickly Acacia/Keekar
- Coral Tree
- Deodar Cedar
- DalbergiaSissoo /Sheesham Tree
- Calotropisprocera/Giant milkweed
- Alovera
- Marijuana

61. Gene Flow:

"The introduction of genetic material (by interbreeding) from one population of a species toanother" **Vertical Gene transfer:** "The transfer of genes from parents to offspring." Horizontalgene transfer:

"**Horizontal gene transfer** is known to occur between different species, such asbetween prokaryotes and eukaryotes, between the three DNA-containing organelles ofeukaryotes, the nucleus, the mitochondrion and the chloroplast.

62. Effects of migration 2

To secure a better climate for living by avoiding unfavorable climatic conditions • e.g. intense cold, hot, stormy conditions and food storage

By alternatively exploiting two different habitat for food due to migration more birds can be able to exist.

Change in climate provide greater variety in birds diet

63. Processing of sample in dna bank 5

DNA banks can now be considered as a means of complimentary conservation. DNA storage is particularly useful for those species that cannot be conserved in traditional seed or field genebanks and nor conserved in situ due to high risk in that area.

Advantages

DNA banking is an efficient, simple and long-term method to conserve the genetic information.

Disadvantages

There are problems with subsequent gene isolation, cloning and transfer of DNA back to aplant and it currently does not allow the regeneration of the same genotype as the originalsample. Storage strategy Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Long-term needs and expected volume and number of samples to be stored will determine organization and repository design.

Processing of samples

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of

the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences.
The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA. S torage Once extracted DNA is a stable biomolecule, although it can easily be degraded during extraction and storage. Quality declines within days in hydrated samples held at room temperature or in

64. Describe reservoirs for biological control agent 3

In the case of classical biological control agents, the genetic diversity of introduced populations may be relatively low because the introduction was based on a small founder population. This lack of diversity may inhibit the ability of the population to respond to climate change.

65. Features of habitat species management area 5

Protection of particular species

✓ Protection of habitats

✓ Active management to maintain target species

✓ Active management of culturally-defined ecosystems

- ***Protection of particular species*****:** to protect particular target species, which will usually be under threat (e.g., one of the last remaining populations);

- ***Protection of habitats*****:** to maintain or restore habitats, which will often be fragments of ecosystems;

- ***Active management to maintain target species*****:** to maintain viable populations of particular species, which might include for example artificial habitat creation or maintenance (such as artificial reef creation), supplementary feeding or other active management systems;

- ***Active management of natural or semi-natural ecosystems*****:** to maintain natural or semi-natural habitats that are either too small or too profoundly altered to be self-sustaining, e.g., if natural herbivores are absent they may need to be replaced by livestock or manual cutting; or if hydrology has been altered this may necessitate artificial drainage or irrigation;

- **Active management of culturally-defined ecosystems:** to maintain cultural management systems where these have a unique associated biodiversity. Continual intervention is needed because the ecosystem has been created or at least substantially modified by management. The *primary aim* of management is maintenance of associated biodiversity.

66. Types of gene bank 2

Such gene bank is of two types: In vivo Gene Bank and In vitro Gene Bank

(i) In vivo GeneBank:

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources

(ii) In vitro Gene Bank:

3. This approach includes the conservation of genetic resources by non-conventional methods. In this approach explants are grown on medium.

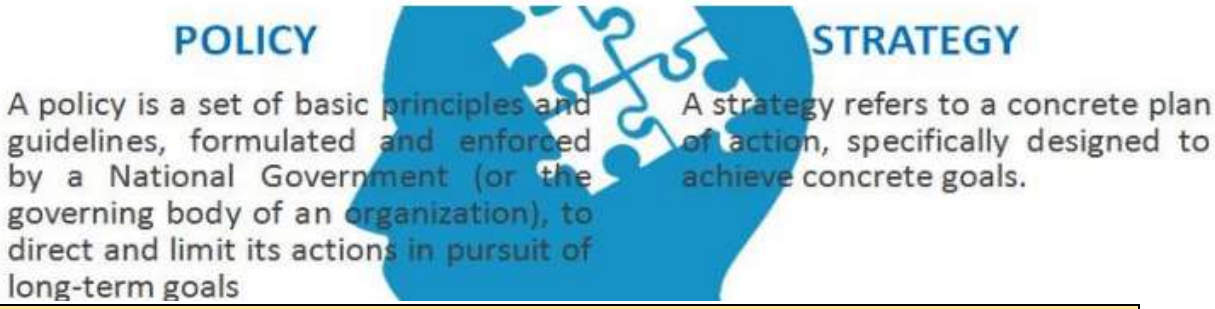
67. Conservation and utilization of AnGR 3+5

Following strategies should be followed for effective conservation and utilization of AnGR:

1. Formulating the National Livestock and wild-life Breeding Policies.
2. Encouraging the Formation of Breed Associations.
3. Developing Professional Human Resources.
4. Strengthening Research and Development Institutions
5. Developing Infrastructure for marketing International co-operation and assistance is needed

in capacity building to remodel available livestock farms/research stations to conserve and develop genetic resources

68. Policy ,strategy????



69. Invasive species

Alien invasive species have had severe impacts on local aquatic flora and fauna, and can upset the natural balance of an ecosytem. For example, the introduction of Nile perch to Lake Victoria has pushed many of the lake's native cichlid species to extinction.

70. Cold storage .

Germplasm of some plants (in the form of shoot tips, nodal or meristem explant culture) are stored at low and non-freezing temperature (1-9°C). At low temperature, growth of plant material is slow down but not completely stopped

as in cryo-preservation. In cold storage there is no risk of cold injuries.

71. Category V details .???

Definition: A protected area where the interaction of people and nature over time has produced an area of distinct character with significant ecological, biological, cultural and scenic value.

Primary objective: To protect and sustain important landscapes/seascaps and the associated nature conservation

Characteristics

- Recognition by artists of all kinds and in cultural traditions (now and in the past);
- Potential for ecological and/or landscape restoration

72. Protected area.2

Protected areas that are strictly set aside to protect biodiversity where human visitation, use and impacts are strictly controlled to ensure protection of the conservation values.

73. Storage strategy

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Longterm needs and expected volume and number of samples to be stored will determine organization and repository design

74. Flowering and Non flowering.

Flowering plants grow flowers and use seeds to reproduce, or make more plants like them. **Nonflowering** plants do not grow flowers, and use either seeds or spores, which are very tiny parts of a plant that can be used to reproduce, to grow more plants just like them.

75. Seed

A seed is an embryonic plant enclosed in a protective outer covering. The formation of the seed is part of the process of reproduction in seed plants, the spermatophytes, including the gymnosperm and angiosperm plants.

76. Pakistan terrestrial water and maritime zones Act 1976.

- It includes provisions for preservation, development and protection of marine environment
- This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
- This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

77. how animal live and move away

Animals need food, water, shelter, and space to survive. Herbivores can live only where plant food is available. Carnivores can live only where they can catch their food. Omnivores can live in many places because they eat both plants and animals.
Animals move for a variety of reasons, such as **to find food, a mate, a suitable microhabitat, or to escape predators**. For many animals, the ability to move is essential for survival and, as a result, natural selection has shaped the locomotion methods and mechanisms used by moving organisms.

78. factors of migration

Multiple factors can drive animal migration, including movement to hospitable environments when local conditions become unfavourable (such as to reduce nutritional and thermoregulatory stress), movement to find mates and/or breeding sites, and movement to minimise competition, predation, infection or parasitism.

79. clonal resporisty

Clonal repository is field Gene bank where genetic resources of clonally propagated crops like fruits are preserved as living plants.
Various institutions are involved in the capacity building to develop AnGR, in the country. These institutions have their own set-ups at federal and provincial level.
These institutions include
Ministry of Food
Agriculture and Livestock at federal level
livestock departments at provincial level

80. write down objective of CBD

The three inter-related objectives are:
The conservation of biological diversity;
The sustainable use of its components;
The fair and equitable sharing of the benefits arising out of the utilization of genetic resources, including by appropriate
- Access to genetic resources,
- Transfer of relevant technologies,
- Funding.

81. Which strategy stored in DNA bank

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Longterm needs and expected volume and number of samples to be stored will determine organization and repository design.

82. endangered specie 2

Leopard,Snow leopard, Indian wild ass, Goitered gazelle

83. treaty 2 marks

Treaty, a binding formal agreement, contract, or other written instrument that establishes obligations between two or more subjects of international law”

- Treaties do not need to follow any special form.
- A treaty often takes the form of a contract, but it may be a joint declaration or an exchange of notes.

84. Cryopreservation

The introduction of cryopreservation technology leads a major breakthrough in animal husbandry .Since the 1st successful cryopreservation of bull semen has been used to propagate the rare and endangered species using assisted reproduction techniques.

85. game reserve

“A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis”

A game reserve (wildlife preserve) is a large area of land where wild animals live safely or are hunted in a controlled way for sport.

86. criteria of national park

National park is an area which is strictly reserved for the betterment of the wildlife & biodiversity, and where activities like developmental, forestry, poaching, hunting and grazing on cultivation are not permitted”

- In these parks private ownership rights are not allowed.
- Their boundaries are well marked and circumscribed.
- They are usually small reserves spreading in an area of 100 sq. km. to 500 sq. km.
- In national parks, the emphasis is on the preservation of a single floral or faunal species

87. Seasonal migration

Definition. **The periodic movement of a population from one region or climate to another in accordance with the yearly cycle of weather and temperature changes**.

88. Genetic variation

Genetic variation refers to **diversity in gene frequencies**. Genetic variation can refer to differences between individuals or to differences between populations. Mutation is the ultimate source of genetic variation, but mechanisms such as sexual reproduction and genetic drift contribute to it as well.

89. Location of cholistan ramsar???

90. Zoological garden

A zoological park is a location where animals are kept in captivity for study and viewing.”

- The zoo originally evolved from the menageries of the ancient world, in which

royalty would exhibit their collection of exotic pets.
 Unfortunately not all zoos are scientific in nature, and extreme controversy has arisen regarding how the animals are treated.
 Suffice to say, regulation is necessary to ensure proper care.
 Conservation (not exploitation) should always be the central goal behind any legitimate zoo.

91. Germplasm conservation techniques application

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed too readily as a basic tool in maintaining genetic diversity. Conventional seed storage offers several advantages:
(1) seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation
(2) good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes;
(3) seed storage is a relatively cheap method for conserving a broad range of germplasm
(4) large land areas are not tied up in conservation
(5) international exchange of genetic material is facilitated by seed storage.

92. define protected areas

Protected areas that are strictly set aside to protect biodiversity where human visitation, use and impacts are strictly controlled to ensure protection of the conservation values.

93. services provided by pepa?

PEPA provides for;
• Protection,
• Conservation,
• Rehabilitation and
• Improvement of the environment;
 PEPA provide framework for prevention and control of pollution
 Helps in protection of sustainable development..

94. 6) limitation of germplasm

seeds of many important tree species cannot adequately survive long-term storage;
(2) seeds of many tropical hardwoods cannot survive even a year of conventional seed storage;
(3) some genetic damage or change in gene frequencies may occur during seed storage. Cryogenic storage at its current level of technology offers great promise for long-term storage of true orthodox and sub-orthodox seeds. It has potential for application to recalcitrant species, but much more research and development will be required.

95. 1) reason for migration salmon

The ultimate purpose for salmon to return to their home streams and rivers is **to reproduce and ensure the survival of their offspring**.

96. assistant migration

Assisted migration is the act of moving plants or animals to a different habitat. The destination habitat may or may not have once previously held the species; the only requirement is the destination habitat must provide the bioclimatic requirements to support the species

97. what types of genetic material are used in DNA bank

DNA banking is the secure, long term storage of an individual's genetic material. DNA is most commonly extracted from blood, but can also be obtained from saliva and other tissues. DNA banks allow for conservation of genetic material and comparative analysis of an individual's genetic information.

98. cryopreservation for viability of cell

- There is possibility of death of cells due to storage stress.
- Thus viability can be found at any stage.
- It is calculated by formula :

$$\left(\frac{No\ of\ cells\ growing}{No\ of\ cells\ thawed}\right) \times 100$$

99. disadvantage of DNA ban

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

100. distinguish features of protected landscape and sea caps

Protecting some of the earth's richness that will not survive outside.

• Protecting additional ecosystem services.

• Providing areas where ecosystems can be studied in as pristine an environment as possible.

101. services provided by the PEPA???

Protection, • Conservation, • Rehabilitation and • Improvement of the environment; • PEPA provide framework for prevention and control of pollution • Helps in protection of sustainable development.

102. explain two types ex situ conservative

(i) In vivo Gene Bank

(ii) In vitro Gene Bank

103. habitat management features...

Habitat management means management of land and its associated resources/features to maintain species in suitable habitats within their natural geographic distribution so that isolated subpopulations are not created. This does not simply maintaining all habitat or individuals of all species in all cases.

104. merits and demerits of in situ conservation

In situ conservation

Advantages

1. Plants and animals conserved in their natural environment
2. Biodiversity permanently protected
3. Representative examples of ecosystems also permenantly protected
4. Natural and cultural heritage protected permenantly
5. Ecological integrity is maintained and managed
6. Opportunities may arise for ecologically sustainable land uses (which come with associated economic benefits)
7. Facilitates scientific research of the site
8. It may be possible to improve the ecological integrity of the area and restore it if it has been damaged by poaching etc.

Disadvantages

- Endangered habitats may be fragmented so the area may not be large enough to ensure the survival of these species
- Genetic diversity may have already been dramatically decreased

- Conditions that threatened the organisms in the area may still be present, e.g. disease or interspecific competition
- Poachers and ecotourists may see the thriving area as an opportunity and may cause damage

105. issue of category of V

Being a relatively flexible model, category V may sometimes offer conservation options where more strictly protected areas are not feasible.

Category V protected areas can seek to maintain current practices, restore historical management systems or, perhaps most commonly, maintain key landscape values whilst accommodating contemporary development and change: decisions about this need to be made in management plans.

The emphasis on interactions of people and nature over time raises the conceptual question for any individual category V protected area: at what point on the temporal continuum should management focus?

106. Orange

Pakistan is the sixth largest producer of Kinow (mandarin) and oranges in the world, with 2.1 million tons The soil and climatic conditions in Pakistan have given the Kinnow a unique flavor.

107. Pakistan terrestrial water and maritime zoon act?

Ans: It includes provisions for preservation, development and protection of marine EnvironmentThis law controls marine pollution and exploration, development, conservation andmanagement of living resources in Pakistan's Exclusive Economic Zone (EEZ)This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

108. Nut Trees in pakistan

Nut Trees in Pakistan are as follow
• Almonds • Walnut • Tree Nuts
Almonds Almonds fall in the family of peaches, plums and nectarines. Seeds of these plants are edible as Almond. Almond plants blossom into beautiful pink flowers in spring in addition to yielding fruits in summer.

109. Anthropogenic activities effecting Genetic Diversity

Any change in the environment, natural or human induced causes a selection of events that only the fittest survive. Anthropogenic impact is apparent in the coastal zone . Man made activities increases the number of changes occurring to individual and populations. Such pressure is exerted by:

• Artificial selection (harvesting, aquaculture) • Degradation of habitats • The release of farmed fish into the wild

110. Why to prevent the loss of Genetic Diversity?

The loss of genetic diversity is difficult to see or measure. In contrast, the reduction and extinction of populations is far easier to see.

• Extinction is not only the loss of whole species, but is also preceded by a loss of genetic diversity within the species.

• This loss reduces the species ability to perform its inherent role in the whole ecosystem.

• The loss of genetic diversity within a species result in the loss of useful and desirable traits (e.g. resistance to parasites).

• Reduced diversity eliminate options to use untapped resources for food production, industry, medicine.

111. Role of national park in the landscape/seascape

• Protecting some of the earth's richness that will not survive outside.

• Protecting additional ecosystem services.

• Providing areas where ecosystems can be studied in as pristine an environment as possible.

• Protecting natural sites that are also of religious and cultural significance.

112. Three Name Research botanical gardens in pakistan

• Abdul Wali Khan University Botanical Garden, Mardan

• Botanical Garden, Governor's House, Lahore

• Botanical Garden, Govt Zamindar College, Gujrat

• Danishmandan Botanic Garden, Lahore

113. Cholistan desert location

It is part of the Cholistan desert in the south eastern portion of the province of Punjab. It contains some of the most rare and interesting wildlife in Pakistan.

114. Huntington's Disease

Huntington's disease (aka Huntington's chorea) is a genetic disorder which results in slowly progressing brain cell death. There are two distinct populations in which the disorder occurs much more often.

The first group is the Afrikaner population of South Africa.

. The second group is the residents of the Lake Maracaibo region of Venezuela.

115. Plant genetic resources

Plant Genetic Resources for Food and Agriculture (PGRFA) are the raw material that farmers and plant breeders use to improve the quality and productivity of crops. They can be defined as any genetic material of plant origin of actual or potential value for food and agriculture, e.g. seeds, tubers, mature plants etc.

116. Seasonal migration

Response to change in the season tropical & sub tropical countries area, this occurs in the beginning or end of the warm season. This type of migration is for food or breeding . They migrate from the south to the north during summer known as "summer visitors". e.g. snow bunting, red wing, shore lark